Enrollment No._____

GANPAT UNIVERSITY

M. Sc.IT (MACHINE LEARNING) SEMESTER: III EXAMINATION (CBCS)

P83A1NLP NATURAL LANGUAGE PROCESSING

Nov - Dec 2022

Time: 3 Hours] **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. Answer the following.(Any Six Out of Eight) **(30)**

I Explain the different levels of knowledge used in NLP with example.

II What is meant by the semantics of a natural language? Explain with some example.

III Explain POS tagging with an example.

IV What is the tokenization in NLP? Why do we need tokenization?

V What is Text Mining? What is the difference between text mining and NLP?

VI Why and how is a n-gram useful in NLP?

VII What is a corpus? Give few examples of corpus.

VIII Explain the step involved converting text into features..

2. Answer the following.(Any Two) **(08)**

I What is frequency distribution? How the class FreqDist works?

II What is bigram and trigram in NLP? Explain with some example.

III What are POS tags in NLP? What is the purpose of using Named Entity Recognition (NER)?

3. A. Answer the following.(Any One) **(06)**

I Write the short note on Term Frequency and Inverse Document Frequency (TF-IDF).

II What is the confusion matrix? Explain with some example.

B. Choose the correct answer from the following choice **(01)**

What is Machine translation?

(A) Converts one human language to another

(B) Converts Machine Language to human language

(C) Converts human Language to Machine language

(D) Converts any language to English

Enrollment No._____

GANPAT UNIVERSITY

MSc IT (AI & ML) SEM-III EXAMINATION

P83A2ML2: MACHINE LEARNING-II

[Time: 3 Hours] Nov - Dec 2022 [Total Marks: 60]

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. Answer the following: (ANY SIX OUT OF EIGHT) **(30)**

A. Why clustering is used in machine learning? Explain in brief.

B. Explain k-means clustering algorithm in detail.

C. What is overfitting? Explain the technique that is used to solved the overfitting problem.

D. What is variance? Explain voting-based ensemble learning in brief.

E. Write a short note on recommend system.

F. Explain collaborative filtering system in detail.

G. Explain the DBSCAN clustering algorithm with the figure in detail.

H. Write a short note on semi-supervised learning.

2. Answer the following: (ANY ONE) **(06)**

A. What is dendrogram? Explain Mini-batch k-means clustering algorithm.

B. List down the linkage methods in hierarchical clustering. Explain any three methods in brief.

3. Answer the following: (ANY ONE) **(09)**

A. Write down the diff. between the bagging and boosting.

B. What is bias? Explain bagging ensemble learning with proper example.

C. Write down the steps for performing the AdaBoost algorithm.

4. Answer the following: (ANY ONE) **(06)**

A. Write down the diff. between the collaborative and content-based filtering.

B. What is content-based filtering technique? Explain in detail.

5. Answer the following: (ANY TWO) **(09)**

A. Explain label propagation algorithm in detail.

B. Write down the diff. between the supervised and semi-supervised learning.

C. Explain the types of real-world applications of semi-supervised learning.

END OF PAPER

Seat No._____

GANPAT UNIVERSITY
M.Sc.IT(AI&ML) SEM-III EXAMINATION

P83A3DL-REINFORCEMENT LEARNING & DEEP LEARNING

Time: 3 Hours] Nov - Dec 2022 **[Total Marks: 60**

Instructions:

1. Figures to the right indicate full marks
2. Be precise and to the point in your answer

1. Answer the following: (ANY SIX OUT OF EIGHT) (30)

A. What is Reinforcement learning? Explain the key features of Reinforcement Learning.

B. Discuss the terms used in Reinforcement Learning.

C. Explain how does Reinforcement Learning Work?

D. How bell-man equation is used in RL to find optimal policy?

E. List out an activation functions used in ANN and explain any 2 in details.

F. What is the role of optimizers in ANN? List down all and explain any three optimizers in details.

G. Difference between TensorFlow and Keras libraries.

H. Explain architecture of CNN with detailed description of convolutional layer and max pooling layer.

2. Answer the following: (ANY ONE) (06)

A. Discuss the approaches to implement Reinforcement Learning.

B. Explain the elements of Reinforcement Learning.

3. Answer the following: (ANY TWO) (09)

A. Difference between Reinforcement Learning and Supervised Learning.

B. Explain Q-learning algorithm in detail.

C. Explain how Markov decision problems is used in Reinforcement Learning.

4. Answer the following: (ANY ONE) (06)

A. How to decide which activation function should be used in output layer?

B. What is ANN? Explain the architecture of an artificial neural network.

5. Answer the following: (ANY TWO) (09)

A. What is Deep Learning? Why Deep Learning?

B. Explain RNN algorithm in brief.

C. Explain Auto-encoders algorithm in brief.

END OF PAPER